

Econometric I



$$y_i \quad x_{1i} \quad x_{2i} \quad \dots \quad x_{ki} \quad i=1 \dots n$$

$$E(y_i | x_{1i} \dots x_{ki}) = E[y_i | x_{1i} = x_{1i}^*, x_{2i} = x_{2i}^*, \dots x_{ki} = x_{ki}^*]$$

regression func.

$$\rightarrow x_{1i}\beta_1 + x_{2i}\beta_2 + \dots x_{ki}\beta_k \rightarrow x_i^T \beta$$

$$\begin{bmatrix} \vdots & \vdots & & \\ x_{1i} & x_{2i} & \dots & x_{ki} \\ \vdots & \vdots & & \end{bmatrix}$$

$$\begin{bmatrix} x_{11} & x_{21} & \dots & x_{k1} \\ \vdots & - & - & \vdots \\ x_{n1} & - & - & x_{nk} \end{bmatrix}$$

$$E(y|x) = x^T \beta \quad x's \text{ are fixed.}$$

$n \times k$

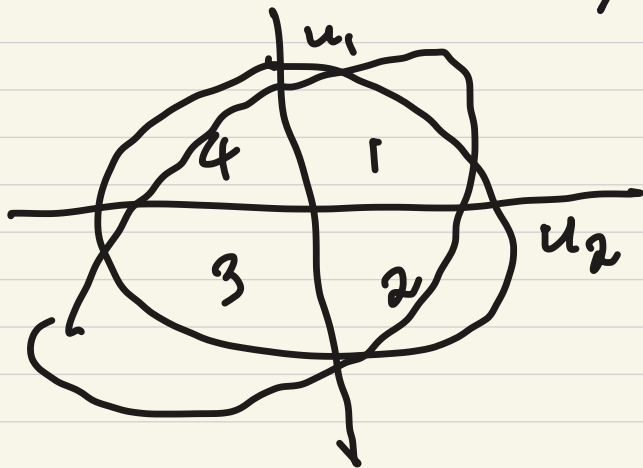
$$y = E(y|x) + \underbrace{y - E(y|x)}_u$$

$$y = E(y|x) + u$$

$$y = X\beta + u \quad \begin{matrix} u \sim N(0, \sigma^2 I_n) \\ n \times 1 \quad \quad n \times 1 \quad \quad n \times n \end{matrix}$$

$$\text{Var}(u) = E(u - E(u))(u - E(u))'$$

$$= E(uu') = \sigma^2 I$$



$$E \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix} \begin{bmatrix} u_1 & \dots & u_n \end{bmatrix}$$

$$= E \begin{bmatrix} u_1^2 & \dots & u_1 u_n \\ \vdots & \ddots & \vdots \\ u_n u_1 & \dots & u_n^2 \end{bmatrix}$$

sphericity assumption

$$\begin{bmatrix} \sigma^2 & 0 & 0 & 0 & \dots \\ 0 & \ddots & & & \\ 0 & & \ddots & & \\ 0 & & & \ddots & \\ 0 & & & & \sigma^2 \end{bmatrix}$$

$$\begin{array}{l} X \rightarrow \text{rank}(X) = k \\ n \times k \end{array} \rightarrow \begin{array}{l} X'X \rightarrow k \times k \\ k \times n \quad n \times 1 \quad \text{rank } k \end{array}$$

$$y = x\beta + u \quad u \sim N(0, \sigma^2 I)$$

$$E(u) = 0$$

$$\begin{aligned} \|u - E(u)\| \quad \|u\| &\rightarrow (u'u)^{\frac{1}{2}} \\ &= \sum_{i=1}^n u_i^2 \end{aligned}$$

$$w \sim N(0, \Sigma_n) \quad \Sigma_n \neq I_n$$

$$\|w\| \rightarrow w' \Sigma^{-1} w$$

$$\min ||u|| = \min_{\beta} (y - x\beta)'(y - x\beta)$$

$$\rightarrow \min_b (y - xb)'(y - xb)$$

$$= (y' - b'x')(y - xb)$$

$$= y'y - b'x'y - y'xb + b'x'xb$$

$$= y'y - 2b'x'y + b'x'xb$$

$$= 2x'y + 2x'xb = 0$$

$$b = (x'x)^{-1} x'y$$

y is random

$$y = x\beta + u \quad u \sim (0, \sigma^2 I_n)$$

$$E(x'u) = 0$$

$$\begin{aligned} \|x'u - E(x'u)\| &= \|x'u\| \Rightarrow (x'u)' [Var(x'u)]^{-1} x'u \\ Var(x'u) &= E[(x'u)(x'u)'] \\ &= E(x'u u' x) = \sigma^2 x'x \end{aligned}$$

$$= \frac{1}{\sigma^2} u' x (x'x)^{-1} x'u$$

$$\min_b (y - xb)' \underbrace{x(x'x)^{-1}x'}_{P_x} (y - xb)$$

$$= \min_b (y - xb)' P_x (y - xb)$$

$$= (y' - b'x') P_x (y - xb)$$

$$= y' P_x y - b' x' P_x y - y' P_x x b + b' x' P_x x b$$

$$\text{FOC} - 2x' P_x y + 2x' P_x x b = 0$$

$$\Rightarrow x' P_x y = x' P_x x b$$

$$b = (x'x)^{-1} x' y$$

$$y = X\beta + u \quad u \sim (0, \sigma^2 I)$$

$$E(X'u) \neq 0 \Rightarrow \begin{matrix} Z \\ n \times p \end{matrix} \quad p \geq k$$

$$E(Z'u) = 0$$